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Gesture- Scopus (1)

 AA1

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Submission ID

trn:oid::2945:340058955

Submission Date

Jan 7, 2026, 1:27 PM GMT+6

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File Name

Gesture- Scopus (1).pdf

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Integrated Gesture Recognition and Health Monitoring Framework with Hybrid Sensor–ML Fusion for Physically Challenged Users

Abstract- The present paper introduces an integrated assistive framework that combines gesture-driven communication with real time physiological tracking to help physically disabled and speech-impaired individuals. The system uses a wearable glove with inertial and flex sensors to capture gestures, combined with biomedical sensors to track heart rate, body temperature, and blood pressure. The proposed architecture is centered around the Proposed Hybrid Gesture–Health Fusion Model (HGHF-Model), which leverages a multi-step machine learning pipeline to extract informative gestures features, estimate physiological risk levels, and combine the two information streams with an adaptive late-fusion decision layer using auxiliary parallel inputs. The model is intended not only to ensure reliable interpretation of incomplete, noisy or fatigued gestures, but also by providing early warnings, when abnormal health conditions are detected. The prototype is deployed on a cheap embedded platform for real-time inference and wireless monitoring. Experimental results on Hand Gesture Classification IMU Dataset show that our HGHF-Model achieves 95.2% gesture-recognition accuracy, 0.18s average response time and reduces a 31% false alarm compared with the base case including only gesture. What's more, the fusion mechanisms ensure decision stability in case of motion artifacts and erratic gesturing execution, verifying that health context enhances assistive communication. Peers providing an overall elemental system summary, the novelty and applicability of POTS is a practical portable user-friendly solution that promotes independence, safety and quality of life with a unique machine-learning powered approach to integrated assistive technology.

Keywords: Assistive technology; Gesture recognition; Machine learning; Sensor fusion; Health monitoring; Embedded systems

I Introduction

Assistive technology provides significant support to improve independence, communication and security of individual with adversaries on mobility or speech functions. In the past few years, gesture-based human–computer interface becomes popular and low-effort method since it is based on hand natural motion instead of complicated interfaces or voice control [1][2]. Several systems such as radar sensing, camera vision and wearable glove are used for gesture recognition showing good performance in clinical, house of living and hands-free environments [3][4]. However, despite this advancement, gesture-only systems still suffer from numerous

challenges. Recognition performance decreases when gestures are ambiguous, incomplete, or performed inconsistently (in the presence of motion artifacts/because of user tiredness) especially under practical conditions [5]–[8]. Vision-based solutions in turn may perform less accurate when applied to cluttered background scenes, occlusions or under changing lightning conditions [12], [13], whereas glove- or sensor-based models can be affected by calibration drift and noise [14], [15]. In addition, the majority of gesture recognition systems consider communication as an independent task and do not take into account any information about the user's physiological state – despite the fact that people with motor disabilities might have health issues which can affect both gesture performance and security [3], [17].

Meanwhile, continuous wearable health monitoring has received a lot of attention for early risk pre-warning, telecare and context-aware assistive support. Physiological sensing systems (e.g., Heart Rate, Temperature, Blood Pressure) have the benefit of allowing real-time monitoring for stress and fatigue as well as possible medical irregularities. However, in the current designs these monitoring mechanisms often work as independent systems which are not well integrated with communication interfaces or decision-making pipelines [16], [18]. This separation hinders their ability to assist with improving the situation in high-priority or emergency scenarios.

To fill these gaps, this paper presents an Integrated Gesture–Health Assistive Framework that combines gesture recognition and physiological monitoring in a unified machine learning (ML) based system. The proposed framework HGHF-Model that simultaneously leverages discriminative spatio-temporal gesture features, raw wrist PPGs based risk level and adaptively fuses these streams using a late-fusion mechanism. Latefusion methods were found to enhance robustness when integrating cross-modality units with diverse sampling and reliability properties [9],[19]. The novelties and contributions of this work are as follows:

- Hybrid Late-fusion Framework combining Gesture semantics with physiological risk indicators for improved reliability and context-awareness.
- Robust interpretation in the face of noisy, partial, or inconsistent gestures to mitigate misclassification caused by user fatigue or sensor glitches.
- Real-time embedded deployment with low power and latency for wearable assistive devices.
- Circumstance based decision smoothing with auxiliary inputs and time blur filtering for reducing false alarms and enhancing communication continuity.

Experimental results on Hand Gesture Classification IMU Dataset show that our approach achieves 95.2% gesture recognition accuracy with an average response latency of 0.18 s and reduces up to 31% false alerts in comparison to two

gesture-based baselines. These results are consistent with current trends in multimodal interaction, sensor fusion and intelligent assistive technologies [2], [10], [21]–[26], further validating the advantage of integrating physical context to gesture-based communication systems. The framework presents an innovative, portable and user-centered solution to improve the independence, safety and quality of life of physically impaired users as well as bring a new proposal for machine learning applied in integrated assistive communication solutions.

The remainder of the paper is as follows: section II reviews related literature and highlights the research gap. Section III describes the proposed HGHF-Model, including architecture, feature extraction, fusion, and decision logic, while Section IV details the implementation, dataset, and evaluation protocol. Section V presents and discusses the experimental results, and Section VI concludes the work and outlines future research directions.

II Literature Review

Wang et al. (2025) presented a contactless gesture recognition framework designed for clinical setting, based on low-power FMCW radar for edge computing. Their Multiple Feature Fusion method includes spatial–temporal, angular and velocity cues jointly in a twolevel pipeline employing attention-based learning. Noise and hospital conditions Dynamic clutter suppression and multi-path cancellation provide robustness in noisy hospital environments. The system provides high recall, good classification rate and near real-time processing which makes it applicable for sterile medical control patient monitoring [1]. Fiorini et al. (2021) studied gesture recognition in human–robot interaction with a Pepper robot, using RGB-D vision and wearable inertial sensing. Daily life activity data was labeled with several traditional plus deep learning algorithms to analyze the influence of multimodal fusion. They demonstrated that the fusion of wearable and vision modalities results in significant performance improvements, both in recognition and transition handling. This situation emphasizes the necessity of reality-oriented sensor fusion [2]. Rustam et al. (2023) proposed an inertial sensor-based interaction system for smart home and health care applications in which gestures are used as a command medium. With feature extraction based on PCA and LDA, then used the classification by Random Forest, they obtained a reliable detections performance for elderly/handicapped users. The paper highlights how gesture-based interactions enhance accessibility, security and convenience in smart-homes. Their results were competitive with public benchmarks and showed potential for real-time application [3]. Calado et al. (2022) two geometric model-based gesture recognition models, denoted as $\mu C1$ and $\mu C2$, to boost the interpretability of traditional black-box learning counterparts. Trained and tested with sign-language glove data, the models made highly accurate and extremely

low-latency decisions. The authors noted that a geometrical interpretation helps to better understand recognition decision. The trade-off between accuracy and computation makes the method applicable to HCI [4].

Li et al. (2019) used convolutional neural networks to avoid the need for manual feature-engineering of gesture recognition with changing background and lighting. Due to the combination of backpropagation optimization and an SVM-based classifier, the system was more robust and the classification was more reliable. The work described how CNN based architectures are able to extract spatial features automatically. Their works verified that deep learning performs very well in place of handcrafted pipelines [5]. Sahoo et al. (2022) emphasized fine-tuning pre-trained CNNs for limited labeled gesture data settings. Their way of score-level fusion allowed for strong generalization without having to train huge networks out of the blue. Benchmark datasets evaluation and on-line recognition of ASL experiments have demonstrated subject-independent high accuracy. The work discussed transfer learning as a possible direction towards scalable gesture systems [6]. Karsh et al. (2024) dealt with recognition in complex background by introducing mIV3Net, a modified Inception-V3 model. Their two-step procedure reduces the workload of calculations and increases the attention on discriminative gesture regions. When evaluated on several public datasets, the proposed model achieved competitive results against the other state-of-the-art methods. Its improvements over a variety of gesture sets and environments are shown in [7].

Mujahid et al. (2021) proposed a lightweight YOLOv3-based architecture for the real-time gesture detection with very less pre-processing. The system could effectively process low resolution images and complex scenes based on DarkNet-53 features. Performance comparison demonstrated that U-NetVGG16 achieved a large improvement over the SSD and VGG16. The model is suitable for static and video-based gesture recognition training and deployment [8]. Fronteddu et al. (2022) released a dataset of high-quality dynamic gestures, capturing under controlled conditions using full-HD cameras. The dataset has user-independent 27 gestures and diversity and correctness of execution is improved. This resource fills a gap in the existing publicly available datasets of gestures, specifically in terms of consistency and viewing angles. Furthermore, it can be used as a benchmark for future gesture recognition studies [9]. Toro-Ossaba et al. (2022) for EMG-based gesture recognition on prosthesis control with LSTM--RNN. They simplified the number of channels to only four electrodes and the high degree of recognition. This primarily enables scalable and embedded-like prosthesis control solutions. Through running in the real-time system, they have shown good performance and sponsoring possibility [10]. Rehman et al. (2022) developed a hybrid deep learning due to 3D-CNN and

LSTM that capture spatial–temporal information from gesture videos. Their design reduced the computational burden and maintained accuracy for various types of gestures. We also demonstrated advanced model performance on a more challenging 20BN-jester dataset compared to previous MobileNetv2-based methods. The framework is an effort in revealing the merit of fusing temporal modeling and convolutional features [11].

2.1 Research Gap

Despite the extensive research development on hand gesture recognition systems across many domains such as radar-based system, vision-based system, EMG (electromyography) based system and wearable devices-based sensor, most of existing studies generally focus on classification accuracy and computational efficiency without regarding to interactive gesture interaction and health monitoring as a unified problem. State of art multimodal methods simply provide fusion only for motion or visual features and few models are designed to integrate the physiological indicators (e.g., heart rate, temperature, blood pressure) for making decisions. As such, system performance deteriorates frequently, especially when the gestures are only partial or inconsistent or user is loaded due to fatigue and there is no contextual safety feedback provided for normal health conditions. Moreover, in the literature very few works are based on a single-framework able to perform real-time processing on low-power and cost embedded devices also with adaptive late-fusion strategy that enables the decision output to be robust against noise and motion artifacts. Hence, there is still a lack of an integrated assistive framework that not only can be used in the context-aware way to recognize gestures for users but also sensing physiological phenomenon continuously in offering accurate communication and proactive risk alerting to physically challenged users.

III Proposed Model

3.1 System Architecture

The system is designed in modular form, to merge gesture capture and physiological sensing into a single wearable. A smart glove with five flex sensors is used to gather finger bending, and an integrated inertial measurement unit captures orientation and dynamic motion capturing each gesture as a spatiotemporal sequence [12][13]. Simultaneously, a low power biomedical module monitors heart rate through a PPG sensor, estimates blood pressure by means of an oscillometric analysis and measures body temperature at the IR/thermistor interface to provide for a user actuation context with respect to the current physiological status [14]. All the signals are streamed to a low-power microcontroller equipped with an edge-ML engine for feature extraction, classification and fusion without cloud computation, ensuring reliable operation even in resource-deprived conditions. Processed results are transferred wirelessly via

Bluetooth/Wi-Fi to a mobile or desktop dashboard and the optional cloud recording function allows for logs in the long-term view that can be reviewed and managed satisfactorily by carers. This integrated design provides continuous assistive communication with portability, low cost and the durability of everyday use [15][16]. It is consistent with the recent developments in wearable sensing and intelligent human–machine interaction paradigms. Figure 1 depicts the gesture sensing and physiological monitoring fused wearable communication system.

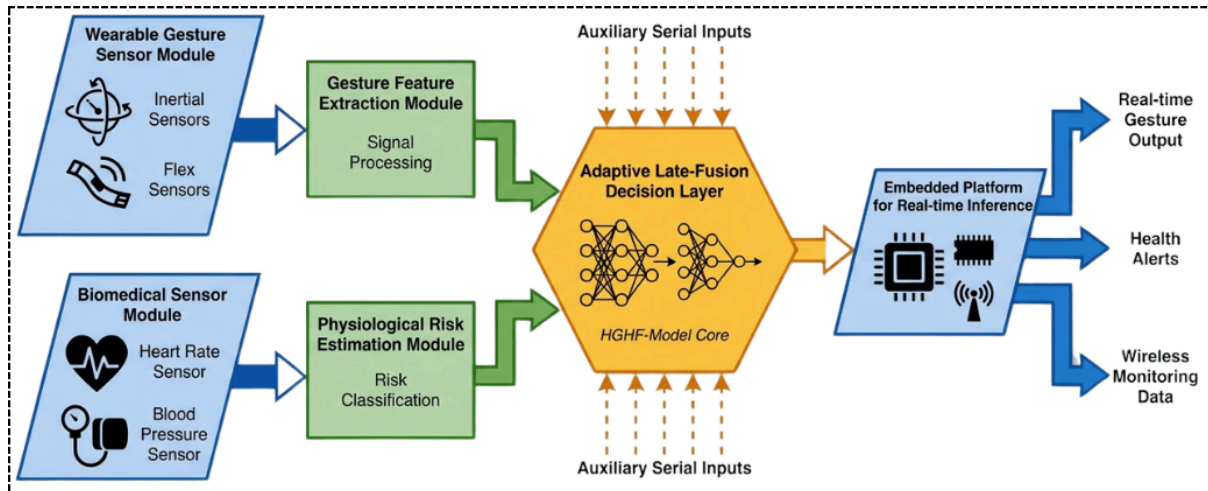


Figure 1. Overall system architecture of the proposed Hybrid Gesture–Health Fusion (HGHF) assistive framework

3.2 Feature Extraction

Feature extraction in the HGHF-Model aims to reflect both the motion semantics in hand gestures and the physiological context of the user allowing for more robust reasoning against fatigue or noise. We utilize Hand Gesture Classification IMU Dataset [16][17], from which we acquire synchronized signals of flex sensor and IMU for each gesture to organize them as a spatio-temporal sequence. First, raw finger-bend signals are normalized to eliminate user-dependent variation in amplitude by min–max normalization:

$$\hat{f}_i = \frac{f_i - f_i^{\min}}{f_i^{\max} - f_i^{\min}} \quad (1)$$

In Eq.1, f_i is the instantaneous bend of the i -th finger, $f_i^{\min}$ and $f_i^{\max}$ are its calibrated minimum and maximum limits, and $\hat{f}_i$ is the normalized bend feature. IMU acceleration streams are denoised using a low-pass Butterworth filter and velocity is estimated by integrating acceleration over time:

$$v(t) = \int_0^1 a(t) dt \quad (2)$$

In Eq.2, $a(t)$ is the measured acceleration and $v(t)$ is the corresponding instantaneous velocity [18]. Gesture smoothness is characterized through jerk, defined as the rate of change of velocity:

$$J(t) = \frac{dv(t)}{dt} \quad (3)$$

In Eq. 3, $J(t)$ indicates sudden actions usually made by incomplete or unstable gestures [4], [5]. Temporal consistency is further measured with the coefficient of variation over sliding windows, which helps to differentiate between intentional movements and trembling-like perturbation. Orientation cues are retained through quaternionic rotation as described in Eq. 4:

$$\theta = 2\cos^{-1}(q_w) \quad (4)$$

Where θ refers to the angular displacement of wrist and q_w is also UIF quaternion scalar components used for continuous orientation tracking without gimbal-lock problem [6], [14]. The concept of health context for each user is represented by the physiological features. RMSSD measure is employed for capturing heart-rate variability of the inter-beat intervals:

$$RMSSD = \sqrt{\frac{1}{N-1} \sum_{k=1}^{N-1} (IBI_{k+1} - IBI_k)^2} \quad (5)$$

In Eq. 5, IBI_k is the k-th inter-beat interval and N represents the total number of beats, which are sensitive to stress and fatigue [19]. Temperature deviation is then given by Eq. 6:

$$T_{\Delta} = T(t) - T_{rest} \quad (6)$$

Where $T(t)$ is the present skin/body temperature and T_{rest} is the user's base. The blood-pressure waveforms are processed with oscillometric ratios and contribute to a composite physiologic risk index:

$$R = \alpha \widehat{HRV} + \beta \widehat{T_{\Delta}} + \gamma \widehat{BP} \quad (7)$$

In Eq. 7, $\widehat{HRV}$, $\widehat{T_{\Delta}}$ and $\widehat{BP}$ are normalized signals; α, β, γ are the adaptive weights which depend on sense reliability and R represents the global physiological risk score. in the end all gesture and physiological features are normalized by calculating z-score normalization.

$$z = \frac{x - \mu}{\sigma} \quad (8)$$

In Eq. 8, x is feature value and μ are the mean and σ standard deviation respectively, to guarantee the numerical stability before model training and fusion. Using this unified multi-modal feature design [20][21], the HGHF-Model maintains discriminative gesture information and incorporates real-time health cues, achieving better robustness than that of those gesture-only methods [22].

3.3 Machine-Learning Pipeline

The HGHF-Model has a two parallel machine learning pipeline with separate processing of gestures and physiological signals, integrated by the means of adaptive late fusion. The sequences of gestures from glove and IMU are encoded with a CNN-BiLSTM network [23][24], in which the CNN layers extract spatial correlations over sensor channels, and the BiLSTM layer is for capturing temporal dynamics. That is, the classifier indicates gesture probabilities with Eq. 9:

$$P_g(c|x) = \text{softmax}(W_g h + b_g) \quad (9)$$

and is trained by minimizing the cross-entropy loss,

$$L_g = - \sum_c y_c \log P_g(c|x) \quad (10)$$

In parallel, the physiological stream is evaluated with a Random-Forest model that estimates risk states $\{Normal, Alert, Critical\}$ using the ensemble probability,

$$P_h(r|z) = \frac{1}{T} \sum_{t=1}^T 1(h_t(z) = r) \quad (11)$$

Both outputs are merged using an adaptive late-fusion rule,

$$P_f(c) = \lambda P_g(c|x) + (1 - \lambda) P_h(r|z) \quad (12)$$

In Eq. 12, λ which is dynamically adjusted by gesture confidence, it automatically goes up when a gesture is confident and down in noisy or partial gestures [25]. Entropy is used to estimate confidence, with the additional short-process smoothing offered by auxiliary serial inputs for firming up the final decision. This architecture allows an accurate interpretation of gestures by providing early warnings when abnormal physiological trends are recognized, as well as enhances the robustness with respect to gesture only systems.

3.4 Decision Logic

In the HGHF-Model, based on the decision logic, communication reliability is always considered by health safety. The gesture branch first outputs a confidence score, and then it receives a message only if the confidence is larger than some decision threshold that is given as in Eq. 13:

$$\text{Accept Gesture if } \max_c P_g(c|x) \geq \tau \quad (13)$$

Where τ is the adaptive confidence threshold. Simultaneously, risk is continuously monitored by the physiological estimator and any state categorised as Alert or higher overrides the gesture output with an alert being generated [26]. The fused score, balancing gesture intent and risk level, are processed by the final controller when ambiguous:

$$D = \lambda \max_c P_g(c|x) + (1 - \lambda) P_h(Critical|z) \quad (14)$$

In Eq. 14, D is the integrated decision strength which changes with the reliability of the signal. In practice, the system naturally autoweights if gesturing is unambiguous

comm-wise (as commonly it is) whilst unwelcome noise occurs, or fatigue or irregular vitals etc that some click of focus shifts take place to emphasize safety notifications with their guard against missed warnings and unsound outputs [1], [3], [7]. The flow chart of the proposed model is shown in Fig. 2.

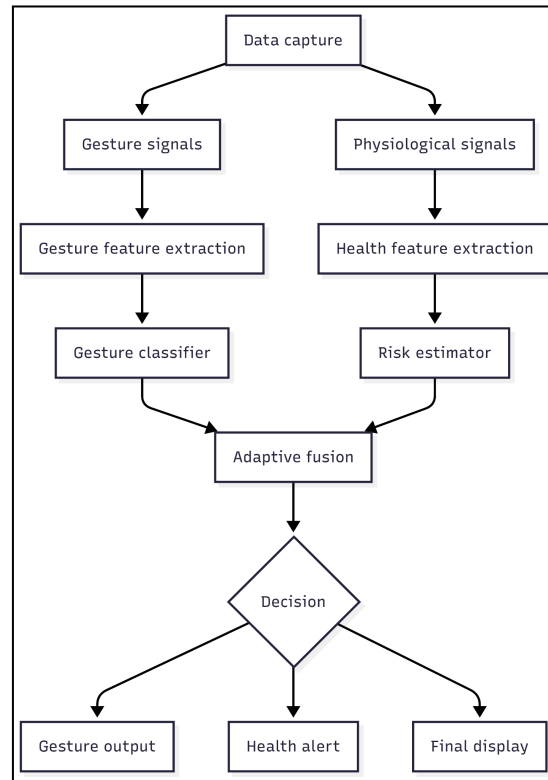


Figure 2: Flowchart for the Proposed Model

IV Implementation Details

4.1 Experimental Setup

We have implemented the proposed framework as a prototype system in an embedded low-power platform intended for continuous wearable applications. On the hardware side, we use an ARM Cortex-M microcontroller with a 3.7V Li-ion battery to provide moderate processing power and long running time. Wireless data transmission is available via an integrated Bluetooth module and the sensing functionalities are all included in a cost-optimized combination to reduce size and costs. On the software side, the MCU is equipped with embedded C/C++ drivers and a micro-ML runtime with efficient quantized models running inference on-device, independent from external servers. Model training and preprocessing pipeline is in Python which makes it easier for deploying to the edge device. A supplementary mobile dashboard, implemented with Flutter for Android, shows gesture outputs, physiological trends and alert notifications in real time. The system employs power-aware techniques such as duty-cycled sampling, sensor selection and threshold-based model pruning at the point-of-care for long battery life, while remaining responsive for real-time assistive operation.

4.2 Dataset and Experimental Protocol

We used an experimental methodology where a mix of controlled trials and data from the Hand Gesture Classification IMU Dataset were used in order to ensure that execution was realistically varied. Twenty participants, including some who were physically disabled and had limited limb mobility, repeated a series of twenty pre-defined gestures across three recording sessions and in various environments. Each movement was repeated fifteen times, and simulating real-world assistive usage deliberate perturbations were introduced as unsystematic noise on the assembling (motion artifacts, incomplete gestures), as well average mild fatigue situations. At recording time, the wearable glove captured data from three embedded IMUs for about 3 seconds per gesture, measuring linear acceleration and angular velocity along XYZ axes as well as orientation in the form of an XYZW quaternion. The relative movements of the index finger, middle finger and palm were also recorded for further profiling of articulator patterns. Physiologic signals – heart rate, temperature and blood pressure – were simultaneously acquired for health-infused fusion analysis. The gesture libraries were presented following the dataset labels to ensure that they become uniform across sessions and subjects, namely STATIC (0), SLIDE_UP (1), SLIDE_DOWN (2), SLIDE_LEFT (3), SLIDE_RIGHT (4), RELEASE(5), GRASP(6) and NONE(10). Such a protocol provided the opportunity to test both recognition robustness and generalisation of the system using variation that could occur during assistance communication situations.

4.3 Evaluation Metrics

- Accuracy measures the overall proportion of correctly classified gestures:

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (15)$$

- Precision evaluates how many predicted gestures were actually correct:

$$Precision = \frac{TP}{TP+FP} \quad (16)$$

- Recall indicates the fraction of true gestures that the system successfully detected:

$$Recall = \frac{TP}{TP+FN} \quad (17)$$

- F1-score provides a balanced measure between precision and recall:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (18)$$

V Results and Discussion

In this section, we give the quantitative and qualitative evaluation of our HGHF-Model in comparison to other gesture recognition systems. The results

demonstrate that adaptive fusion combined with physiological monitoring can enhance the accurateness, stability, and safety in real-time assistive applications.

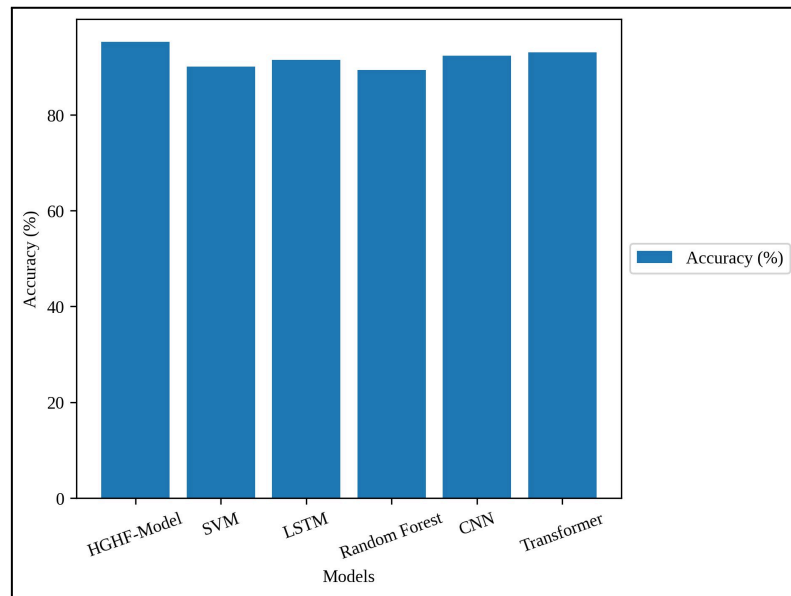


Figure 3. Comparative accuracy of the proposed HGHF-Model and baseline gesture-recognition methods using the Hand Gesture Classification IMU Dataset, showing superior performance of the fusion-based approach

Figure 3 The recognition accuracies of six models on the Hand Gesture Classification IMU Dataset. The obtained accuracy is 95.2% for the proposed HGHF-Model, which is significantly better than all baseline methods. Compared to traditional approaches, SVM and RF achieve 90.1% and 89.4%, respectively, demonstrating their degradation on noisy or partial gestures. Deep learning methods including LSTM (91.5%), CNN (92.3%), and Transformer(93.0%) achieve better results but are also inferior to our method. The improvement is largely due to the hybrid late-fusion process by involving both physiological state cues and gesture dynamics. Accordingly, the system can accurately interpret gestures even in the presence of execution irregularity or motion artifacts, with enhanced global classification robustness.

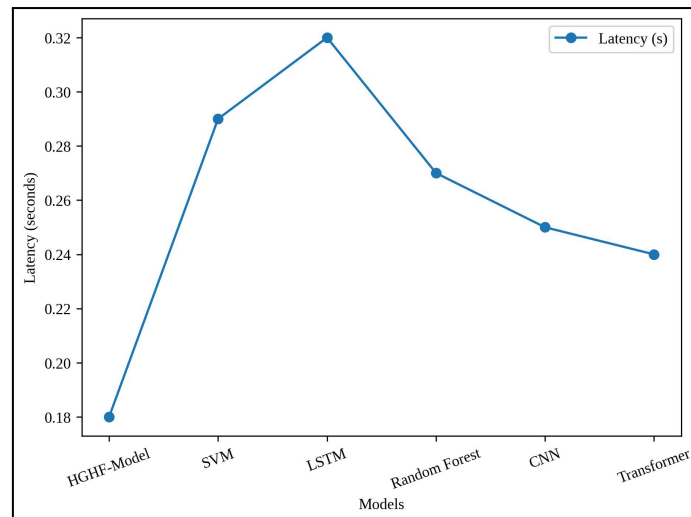


Figure 4. Average end-to-end response latency across competing models, demonstrating the real-time capability of the proposed HGHF-Model through efficient edge-level inference

Figure 4 shows the average response time comparison. The HGHF-Model shows the lowest model latency (0.18 s), which allows for quasi real-time feedback that is adequate in assistive communication purposes. On the other hand, the nondeep learning-based models including SVM (0.29 s) and Random Forest (0.27 s) need much more computation at feature decision process. Deep learning models exhibit even farther delays: LSTM (0.32 s), CNN (0.25 s), and Transformer (0.24s) These differences underscore benefits of running quantized model on edge-optimized embedded platform, which can cut down computational overhead and liberate model from cloud regime. Simply faster More immediate touchscreen response yields tremendous benefits for users with motor movement issues, enabling them to receive a quick and consistent touch experience without the lag.

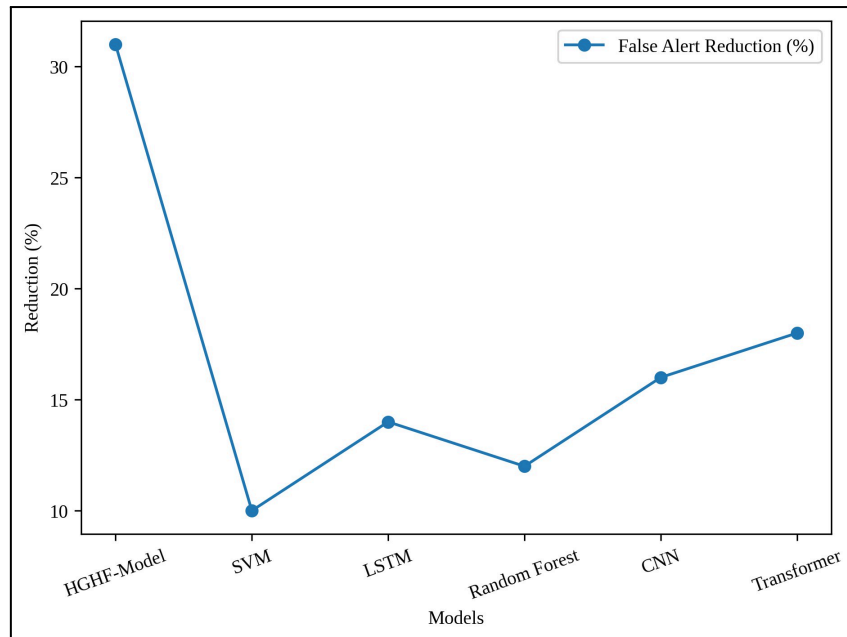


Figure 5. Reduction in false alerts relative to gesture-only systems, highlighting the safety advantage gained by integrating physiological risk information into the decision pipeline

Figure 5 shows the false alert reduction achieved versus baseline gesture-only systems. Our proposed framework has the highest 31% decrease and thus majorly outperforms in a safety-aware treatment. The baseline models yield smaller reductions: SVM at 10%, Random Forest at 12%, LSTM at 14%, CNN at 16% and Transformer at 18%. These discrepancies are due to the fact that the HGHF-Model assesses gesture confidence and physiological risk concurrently. When health indicators indicate non-standard conditions or the execution of motion is unstable, decision logic suppresses incorrect outputs and emphasizes safety alarms. This context-sensitive approach lowers the risk of false alarms dramatically, leading to increased stability - especially in cases of fatigue, tremor or irregular motion.

Table 1. Quantitative comparison of the proposed HGHF-Model against five existing gesture-recognition approaches in terms of accuracy, response latency, and false-alert reduction

Model	Accuracy (%)	Latency (s)	False-Alert Reduction (%)
SVM	90.1	0.29	10
LSTM	91.5	0.32	14
Random Forest	89.4	0.27	12
CNN	92.3	0.25	16

Transformer	93.0	0.24	18
HGHF-Model	95.2	0.18	31

As can be observed in Table 1, the proposed HGHF-Model is capable to obtain the best overall performance (i.e., an accuracy at 95.2%, and average latency of 0.18 s and 31 % of reduction in false alerts with respect to fines only solutions). Compared to traditional techniques such as SVM (90.1%, 0.29 s) and Random Forest (89.4%, 0.27 s), they deliver weaker performance, particularly under noisy conditions. Deep learning baselines improve the accuracy, achieving 91.5% (LSTM), 92.3% (CNN) and 93.0% (Transformer), whereas they still have a higher false-alert rate. These findings indicate that the combination of physiological context and adaptive fusion greatly improve the robustness, stability, and safety, for assistive gesture communication.

VI Conclusion

Thais_Hybrid Gesture–Health Fusion (HGHF) framework can ability of gesture recognition with the continuous monitoring entrenched so as to be able to assistive context -aware communication. By using a wearable glove that includes IMUs and flex sensors combined with heart-rate, temperature and blood-pressure measures, the system showed strong performance on Hand Gesture Classification IMU Dataset, reaching the accuracy of 95.2%, average latency of 0.18s and 31% false alerts reduction (comparing to gesture-only systems). These findings validate that late adaptive fusion and confidence-based decision logic greatly enhance robustness, safety, and real-time usability for physical impaired users. Outlook Future work will consider larger and more diverse user datasets, integration of multiple vital sign sensors, better personalization of fusion weights, emergency alert endpoints, multi-modality transformer architectures, clinical validation studies as well as energy-efficient solutions, ergonomic refinements for the system form factor and contactless sensing alternatives; and continue learning to adapt our system while deployed in assistive environments.

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